

OpenForge

Product Requirements Document (PRD)

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1. Executive Summary

OpenForge is an AI-powered developer platform designed to simplify the process of contributing to open-source software.

The platform analyzes a developer's GitHub profile, understands their technical skills, recommends suitable repositories, identifies beginner-friendly issues, and provides AI-assisted explanations and contribution guidance.

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The objective is to significantly reduce the learning curve associated with open-source contribution while helping developers build stronger GitHub portfolios.

2. Vision

To become the most intelligent developer platform for discovering, understanding, and contributing to open-source software.

Rather than acting as another repository search engine, OpenForge aims to function as an AI mentor that helps developers successfully complete meaningful contributions.

3. Problem Statement

Many developers encounter significant barriers when attempting to contribute to open source.

Common challenges include:

- Finding repositories appropriate for their skill level
- Understanding unfamiliar codebases
- Choosing issues that match their experience
- Navigating large project structures
- Learning contribution workflows
- Building a professional GitHub profile

Existing solutions expose repository data but provide little guidance on where developers should begin or how they should approach contributions.

4. Product Goals

Primary Goals

- Simplify open-source onboarding.
- Reduce repository learning time.
- Personalize repository discovery.
- Recommend suitable contribution opportunities.
- Increase successful pull requests.

Secondary Goals

- Encourage consistent contributions.

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- Improve developer confidence.
- Track contribution growth.
- Build long-term developer portfolios.

5. Target Audience

Primary Users

- College students
- Beginner developers
- Self-taught programmers
- First-time open-source contributors

Secondary Users

- Professional developers
- Technical mentors
- Open-source maintainers
- Coding bootcamp participants

6. User Personas

Persona A — Student

Objective

Contribute to GitHub projects but lacks practical experience.

Pain Points

- Doesn't know where to start.
- Finds repositories overwhelming.
- Cannot identify beginner-friendly issues.

Persona B — Intermediate Developer

Objective

Find projects aligned with current technical skills.

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Pain Points

- Too many repositories.
- Unsure which projects provide meaningful experience.
- Wants contributions that improve career prospects.

Persona C — Maintainer

Objective

Receive higher-quality pull requests.

Pain Points

- Contributors lack project understanding.
- Frequent duplicate or incomplete pull requests.
- High onboarding effort.

7. User Journey

Landing Page

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GitHub Authentication

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Profile Analysis

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Skill Extraction

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Repository Recommendations

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Repository Exploration

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Issue Recommendation

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AI Repository Explanation

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Contribution Planner

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Pull Request Submission

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Contribution Dashboard

8. Functional Requirements

8.1 Authentication

Features

- GitHub OAuth
- User profile
- Session management
- Secure authentication

8.2 GitHub Profile Analysis

Collect:

- Languages
- Frameworks
- Public repositories
- Commit activity
- Stars
- Forks
- Contribution graph

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- Repository topics

8.3 Skill Analysis Engine

Generate:

- Primary programming languages

- Framework preferences
- Experience estimation
- Skill score
- Technology recommendations

8.4 Repository Recommendation Engine

Recommend repositories based on:

- Programming language
- Technology stack
- Skill match
- Difficulty
- Repository activity
- Community size
- Repository health

8.5 Repository Explorer

Display:

- Repository overview
- README summary
- Technology stack
- Stars
- Forks
- Contributors
- License
- Releases
- Branches

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8.6 AI Repository Understanding

Generate:

- Plain-language repository summary
- Architecture overview

- Folder explanations
- Entry point identification
- Important files
- Dependency relationships

8.7 Issue Recommendation Engine

Recommend issues using:

- Skill compatibility
- Difficulty level
- Labels
- Good First Issue
- Help Wanted
- Estimated effort
- Activity level

8.8 AI Issue Explanation

Generate:

- Simplified explanation
- Required knowledge
- Relevant files
- Suggested implementation approach
- Expected learning outcome
- Estimated completion time

8.9 Personalized Learning Roadmap

Provide:

- Current skill assessment
- Missing technologies
- Learning sequence
- Suggested repositories

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- Recommended contribution milestones

8.10 Contribution Dashboard

Track:

- Pull requests opened
- Pull requests merged
- Issues solved
- Repository contributions
- Programming languages
- Contribution streak
- Growth statistics

8.11 Notifications

Notify users about:

- New repository recommendations
- Matching issues
- Pull request updates
- Maintainer responses
- Trending repositories

9. Non-Functional Requirements

Performance

- Landing page under 2 seconds
- Repository analysis under 15 seconds
- AI responses under 10 seconds

Security

- OAuth authentication
- Secure token management
- HTTPS
- Encrypted credentials

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Scalability

- Support 100,000+ users

- Horizontal backend scaling
- Repository caching

Availability

Target uptime:

99.9%

10. AI Capabilities

The AI system will assist users by providing:

- Repository summaries
- Architecture explanations
- Folder explanations
- Issue explanations
- Learning recommendations
- Contribution planning
- Beginner guidance

11. Dashboard Modules

The dashboard will contain:

- Home
- Repository Recommendations
- Repository Explorer
- Recommended Issues
- Learning Roadmap
- Contribution Dashboard
- Notifications
- Profile
- Settings

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12. Technical Stack

Frontend

- Next.js

- React
- TypeScript
- Tailwind CSS
- shadcn/ui
- Framer Motion

Backend

- Node.js
- Express.js

Database

- PostgreSQL
- Supabase

Authentication

- Supabase Authentication
- GitHub OAuth

APIs

- GitHub REST API
- GitHub GraphQL API

Artificial Intelligence

- OpenAI
- Gemini
- Ollama (optional local inference)
- Retrieval-Augmented Generation (RAG)

Deployment

- Vercel
- Docker

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13. Success Metrics

The following KPIs will measure product success:

- Daily active users
- Weekly active users

- Repository recommendation click-through rate
- Pull requests created
- Pull requests merged
- Repository onboarding time
- AI explanation satisfaction
- User retention

14. Future Scope

The following features are planned for future versions:

- VS Code Extension
- Browser Extension
- AI Pull Request Review
- AI Codebase Visualization
- Team Workspaces
- Organization Dashboards
- Maintainer Portal
- Contribution Certificates
- Hackathon Mode
- Hacktoberfest Integration
- Community Discussions
- Mobile Application

15. Out of Scope (Version 1)

The following capabilities are intentionally excluded from Version 1:

- Git repository hosting
- Integrated IDE
- CI/CD pipeline management
- Self-hosted Git providers
- Live collaborative coding
- Project management features
- Team chat

Product Vision Statement

OpenForge aims to bridge the gap between developers and open-source communities by combining GitHub data, intelligent recommendation systems, and AI-powered guidance into a single developer-first platform.

Version 1 focuses on delivering an intuitive, production-ready experience that helps users discover suitable repositories, understand unfamiliar codebases, and confidently make meaningful open-source contributions.

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